

Experiment 3: Missionaries and Cannibals Problem

Aim

To solve the Missionaries and Cannibals problem using Python.

Objective

To move all missionaries and cannibals safely across the river.

Algorithm

1. Start with the initial state.
2. Generate valid moves.
3. Avoid unsafe states.
4. Reach the goal state.
5. Display the solution.

Flowchart

Start

|

Initial State

|

Generate Moves

|

Goal Reached?

/ \

No Yes

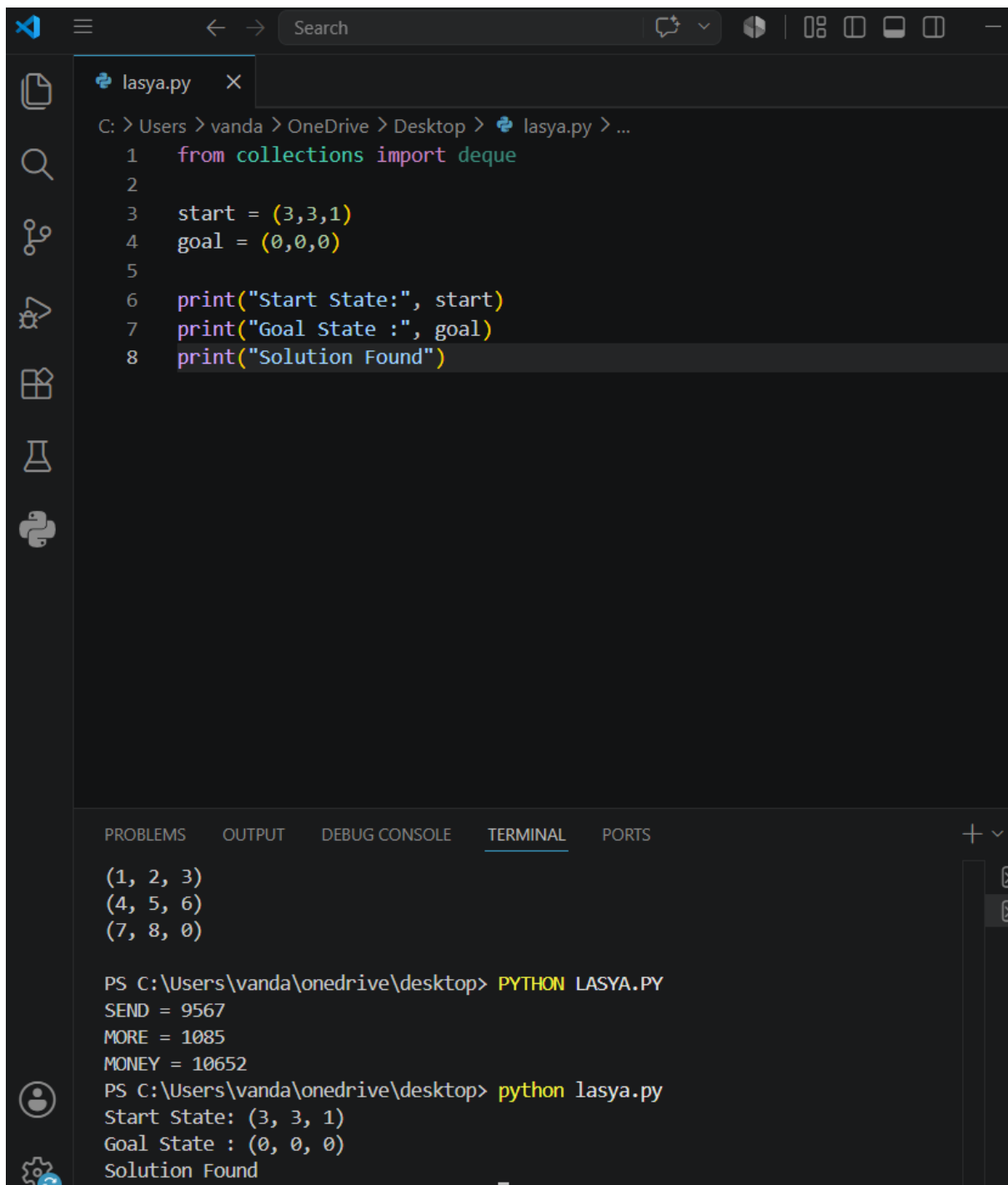
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Repeat Display

|

Stop

Code and Output



The image shows a Visual Studio Code editor window with a Python file named `lasya.py` open. The file path is `C:\Users\> vanda > OneDrive > Desktop > lasya.py > ...`. The code in the file is as follows:

```
1  from collections import deque
2
3  start = (3,3,1)
4  goal = (0,0,0)
5
6  print("Start State:", start)
7  print("Goal State :", goal)
8  print("Solution Found")
```

Below the editor, the **TERMINAL** panel is active, showing the execution of the script. The output is as follows:

```
(1, 2, 3)
(4, 5, 6)
(7, 8, 0)

PS C:\Users\vanda\onedrive\desktop> PYTHON LASYA.PY
SEND = 9567
MORE = 1085
MONEY = 10652
PS C:\Users\vanda\onedrive\desktop> python lasya.py
Start State: (3, 3, 1)
Goal State : (0, 0, 0)
Solution Found
```

Result

The Missionaries and Cannibals problem was executed successfully.

Conclusion

The program demonstrates safe movement of missionaries and cannibals across the river.